UNIVERSITY OF THE ARMED FORCES – ESPE
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36862 – Object-Oriented Programming

UNIT 1 PROJECT · TEAM WORK

Use Case Specifications: ProvStock

Prov&Tec Management, a Comprehensive Sales and Inventory System

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Revision Note

> *Version 1.1 (compact) presents the use case specifications with the same structure as the other documents of the Unit 1 project (Project Definition v1.1 and Product Backlog v1.2). Each use case is summarized in one card with its primary actor, related requirements, description, main flow, and its exceptions and secondary paths in one line each. The complete specification, with preconditions, postconditions, notes, and the detailed alternative flows, is in ProvStock_UseCaseSpecifications_v1.1. The 20 use cases and their main flows are the same in both documents and come from the use case diagram of the Software Requirements Specification (SRS) v1.2.*

1. Introduction

1.1 Purpose

This document specifies, step by step, the 20 use cases of ProvStock, the sales and inventory management system for Prov&Tec. Each specification describes who starts the use case, the interaction between the user and the system, and the errors and secondary paths. It complements the Software Requirements Specification (SRS) v1.2 of ProvStock, whose Figure 1 shows the use case diagram reproduced in Figure 1 of this document.

1.2 Conventions

Every use case is described with the same template (Table 1). The requirement identifiers (FR, NFR, BR, UI) refer to the SRS v1.2.

Table 1
Use Case Template

Field	Content
Use case	Identifier (UC-xx) and name, written as a verb phrase.
Primary actor	The actor who starts the use case. The Owner inherits every use case of the Salesperson and the Warehouse Clerk.
Backlog and requirements	The Product Backlog item and the SRS requirements and business rules that the use case satisfies.
Relationships	«include» (behavior always performed as part of the use case) and «extend» (optional behavior added under a condition).
Description	A short summary of the goal of the use case.
Main flow	The numbered steps of the most common successful path.
Exceptions and secondary paths	Errors and other paths in one line each, referenced by the step where they start (for example, 3a) or by A1, A2… for the secondary options of a “Manage” use case.

1.3 Actors

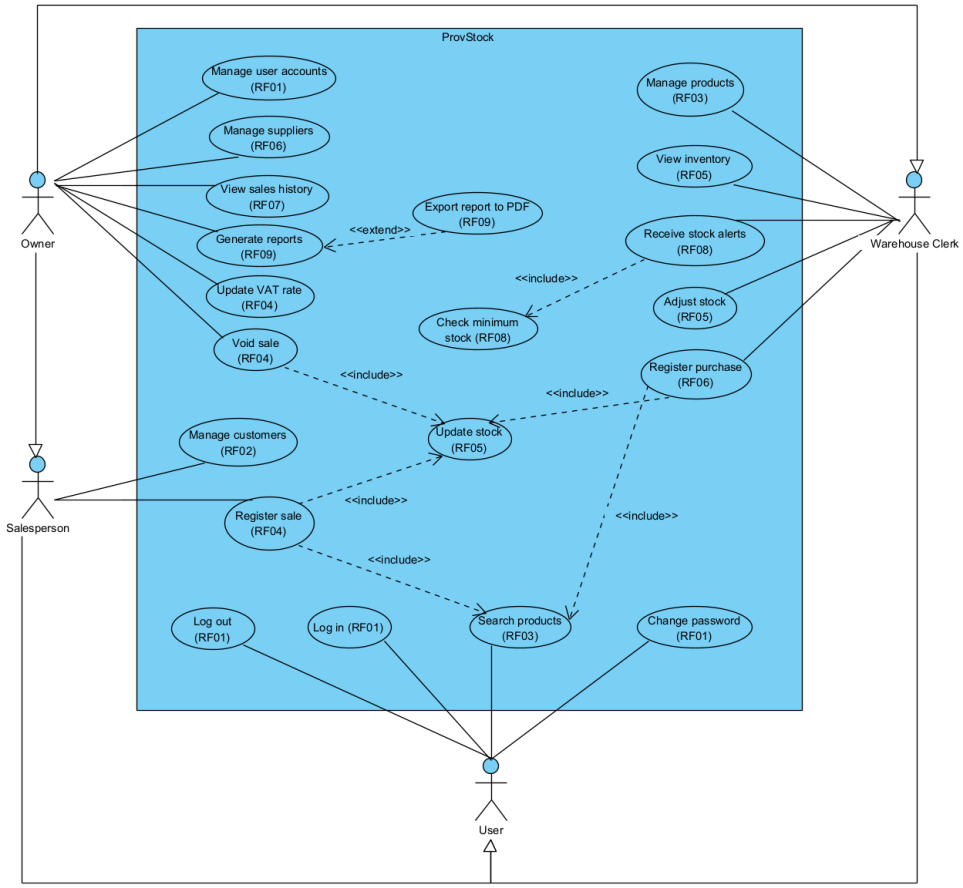
Table 2 describes the actors of the use case diagram (Figure 1).

Table 2
Actors of ProvStock

Actor	Description
User (abstract)	Any person with an account in the system. It groups the functions shared by all roles: logging in, logging out, changing the password, and searching for products.
Salesperson	Serves customers at the counter, registers sales, and registers or updates customers. Specializes User.
Warehouse Clerk	Manages the products and the stock in the warehouse, registers purchases, adjusts the stock, and receives the stock alerts. Specializes User.
Owner	Responsible for the business. Specializes both the Salesperson and the Warehouse Clerk, so it can perform all their use cases; in addition, it manages user accounts and suppliers, voids sales, updates the VAT rate, reviews the sales history, and generates reports.

Update stock (UC-11) and Check minimum stock (UC-17) have no actor of their own: the system performs them automatically as part of other use cases.

Figure 1
Use Case Diagram of ProvStock



Note. Reproduced from Figure 1 of the SRS v1.2. Hollow arrows indicate actor generalization; the Salesperson and the Warehouse Clerk share one generalization toward User. Dashed arrows indicate «include» and «extend» relationships. The codes in parentheses refer to the Product Backlog items.

2. Use Case Summary

Table 3 lists the 20 use cases, their primary actor, the Product Backlog item they come from, and their relationships.

Table 3
Use Case Summary

ID	Use Case	Primary Actor	Backlog	Relationships
UC-01	Log in	User	RF01	—
UC-02	Log out	User	RF01	—
UC-03	Change password	User	RF01	—
UC-04	Manage user accounts	Owner	RF01	—
UC-05	Manage customers	Salesperson	RF02	—
UC-06	Search products	User	RF03	Included by UC-08, UC-15
UC-07	Manage products	Warehouse Clerk	RF03	—
UC-08	Register sale	Salesperson	RF04	«include» UC-06, UC-11
UC-09	Void sale	Owner	RF04	«include» UC-11
UC-10	Update VAT rate	Owner	RF04	—
UC-11	Update stock	— (system)	RF05	Included by UC-08, UC-09, UC-15
UC-12	View inventory	Warehouse Clerk	RF05	—
UC-13	Adjust stock	Warehouse Clerk	RF05	—
UC-14	Manage suppliers	Owner	RF06	—
UC-15	Register purchase	Warehouse Clerk	RF06	«include» UC-06, UC-11
UC-16	View sales history	Owner	RF07	—
UC-17	Check minimum stock	— (system)	RF08	Included by UC-18
UC-18	Receive stock alerts	Warehouse Clerk	RF08	«include» UC-17
UC-19	Generate reports	Owner	RF09	Extended by UC-20
UC-20	Export report to PDF	Owner	RF09	«extend» UC-19

3. Use Case Specifications

Each use case is described below with the template of Table 1.

UC-01. LOG IN	
Primary actor	User (Salesperson, Warehouse Clerk, or Owner)
Backlog and requirements	RF01 · FR-01.1, FR-01.2, FR-01.3, FR-08.1, NFR-02, UI-01, UI-02.

Relationships	None.
Description	The user identifies with a username and password to access the functions allowed for their role.
Main flow	1. The system shows the login window and asks for the username and password. 2. The user enters the username and password and confirms. The password is hidden while it is typed. 3. The system verifies that the username exists, that the account is active, and that the password is correct. 4. The system opens the session and shows the main menu with only the functions allowed for the role of the user. 5. If the role of the user receives stock alerts, the system shows the number of active stock alerts in the main menu.
Exceptions and secondary paths	3a. Wrong username or password, or inactive account: general error message, without saying which data is wrong.

UC-02. LOG OUT	
Primary actor	User
Backlog and requirements	RF01 · FR-01.6.
Relationships	None.
Description	The user closes their session so that no one else can use the system with their account.
Main flow	1. The user selects the option to log out. 2. The system closes the session and shows the login window.
Exceptions and secondary paths	**1a.** A sale or purchase is not confirmed: the system asks whether to discard it.

UC-03. CHANGE PASSWORD	
Primary actor	User
Backlog and requirements	RF01 · FR-01.5, NFR-02.
Relationships	None.
Description	The user replaces their own password with a new one.
Main flow	1. The user selects the option to change the password. 2. The system asks for the current password, the new password, and the confirmation of the new password. 3. The user enters the three values and confirms. 4. The system verifies that the current password is correct, that the new password has at least 8 characters, and that the new password and its confirmation match. 5. The system stores the new password as a salted hash and shows a confirmation message.

Exceptions and secondary paths	4a–4c. Wrong current password, new password with fewer than 8 characters, or confirmation that does not match: error message.

UC-04. MANAGE USER ACCOUNTS	
Primary actor	Owner
Backlog and requirements	RF01 · FR-01.1, FR-01.4, FR-01.7, BR-07, BR-08, NFR-02.
Relationships	None.
Description	The owner creates, updates, and deactivates the user accounts, assigns their roles, and resets forgotten passwords.
Main flow	1. The owner opens the user accounts option; the system lists the accounts with their username, full name, role, and status. 2. The owner selects the option to create an account. 3. The owner enters the username, the full name, the role (Salesperson, Warehouse Clerk, or Owner), and an initial password. 4. The system verifies that the username is unique and that the password has at least 8 characters. 5. The system saves the account as active, with the password stored as a salted hash, and shows it in the list.
Exceptions and secondary paths	**A1–A3.** Update the full name or role; deactivate the account (it stays in the history); reset a forgotten password (FR-01.7). **4a–4b.** Username already exists or password with fewer than 8 characters: error message.

UC-05. MANAGE CUSTOMERS	
Primary actor	Salesperson
Backlog and requirements	RF02 · FR-02.1 to FR-02.6, FR-04.2, BR-07, BR-08, UI-03.
Relationships	None.
Description	The salesperson registers, updates, searches for, and views customers; the owner can also deactivate them.
Main flow	1. The salesperson opens the customers option and selects the option to register a customer. 2. The salesperson enters the identification type and number, the full name or business name, and, optionally, the phone number, the email, and the address (Table B1 of the SRS). 3. The system verifies that the required fields are complete, that the identification number is unique and has a valid length (10 digits for a cédula or 13 for a RUC), and that the email has a valid format. 4. The system saves the customer as active and shows it in the customer list.

Exceptions and secondary paths	A1–A3. Search by identification number or name; update the data (except the identification number); deactivate (Owner only). A4. From a sale: the customer is registered without leaving the sale (alternative flow 2a of UC-08). 3a–3b. Identification number already exists or invalid field: error message. The “Final Consumer” cannot be modified or deactivated.

UC-06. SEARCH PRODUCTS	
Primary actor	User
Backlog and requirements	RF03 · FR-03.4, FR-03.5, UI-04.
Relationships	Included by UC-08 Register sale and UC-15 Register purchase.
Description	The user finds products by code, name, or category and sees their price, stock, and location in the warehouse.
Main flow	1. The user enters a product code, part of the name, a category, or a combination of them. 2. The system shows the matching active products with their code, name, sale price, current stock, and location. 3. The user selects a product to view its complete information.
Exceptions and secondary paths	**2a.** No product matches: message, and the user can search again.

UC-07. MANAGE PRODUCTS	
Primary actor	Warehouse Clerk
Backlog and requirements	RF03 · FR-03.1, FR-03.2, FR-03.3, FR-03.5, BR-07, BR-08, AS-04.
Relationships	None.
Description	The warehouse clerk registers, updates, and deactivates the products sold by the company.
Main flow	1. The warehouse clerk opens the products option and selects the option to register a product. 2. The warehouse clerk enters the code, name, description, category, brand, unit of measure, location in the warehouse, purchase cost, sale price, and minimum stock level (Table B2 of the SRS). 3. The system verifies that the code is unique, that the sale price is greater than zero, and that the purchase cost and the minimum stock level are not negative. 4. The system saves the product as active with a current stock of zero and shows it in the product list.
Exceptions and secondary paths	**A1–A2.** Update the data (except the code and the stock); deactivate a product that is no longer sold (it stays in the history). **3a.** Code already exists or invalid value: error message, the product is not saved.

UC-08. REGISTER SALE	
Primary actor	Salesperson
Backlog and requirements	RF04 · FR-04.1 to FR-04.7, FR-05.1, BR-01, BR-03, BR-05, BR-06, NFR-03, NFR-04, UI-05.
Relationships	«include» UC-06 Search products; «include» UC-11 Update stock.
Description	The salesperson registers a sale to a customer; the system calculates the amounts and decreases the stock.
Main flow	1. The salesperson selects the option to register a new sale; the system records the date, the time, and the logged-in salesperson. 2. The salesperson selects the customer: a registered active customer or the “Final Consumer”. 3. The salesperson searches for a product (UC-06) and enters the quantity. 4. The system verifies that the quantity is greater than zero and not greater than the available stock, and adds a line with the product, the quantity, the unit price, and the line subtotal. If the product is already in the sale, the quantities are added. 5. The system immediately updates the subtotal, the VAT (with the current rate), and the total. Steps 3 to 5 are repeated for each product. 6. The salesperson selects the payment method (cash, bank transfer, or card) and confirms the sale. 7. The system verifies that the sale has one customer and at least one product, assigns the next consecutive number, and saves the sale with the VAT rate applied. 8. The system performs UC-11 Update stock to decrease the stock of each product; the sale and all its stock changes are saved together or not at all. 9. The system shows a confirmation with the number of the sale.
Exceptions and secondary paths	**2a.** Unregistered customer: it is registered without leaving the sale (same data and validations as UC-05). **4a.** Quantity greater than the available stock: rejected, and the available stock is shown. **5a–6a.** A line is removed (totals are recalculated) or the sale is cancelled (nothing is saved). **8a.** Error while saving: neither the sale nor the stock changes are stored (NFR-04).

UC-09. VOID SALE	
Primary actor	Owner
Backlog and requirements	RF04 · FR-04.8, FR-05.3, BR-04, NFR-04.
Relationships	«include» UC-11 Update stock.
Description	The owner cancels a sale registered by mistake; the sale remains in the history and its stock is restored.

Main flow	1. The owner finds the sale by its number or in the sales history and selects the option to void it. 2. The system shows the data of the sale and asks for the reason. 3. The owner enters the reason and confirms. 4. The system changes the status of the sale to “Voided” and records the reason, the date, and the owner. 5. The system performs UC-11 Update stock to restore the stock of each product of the sale; the change of status and the stock changes are saved together or not at all.
Exceptions and secondary paths	1a. The sale is already voided: the option is not available. 3a. Empty reason: the void cannot be confirmed. 5a. Error while saving: nothing changes.

UC-10. UPDATE VAT RATE	
Primary actor	Owner
Backlog and requirements	RF04 · FR-04.9, BR-05.
Relationships	None.
Description	The owner changes the VAT rate applied to new sales when it is modified by law.
Main flow	1. The owner opens the VAT rate option; the system shows the current rate (15% by default). 2. The owner enters the new rate and confirms. 3. The system verifies that the value is a percentage between 0 and 100 and saves it. 4. The system shows a confirmation message.
Exceptions and secondary paths	**2a–3a.** The owner cancels or enters an invalid value: the current rate is kept.

UC-11. UPDATE STOCK	
Primary actor	None (performed automatically by the system)
Backlog and requirements	RF05 · FR-05.1, FR-05.2, FR-05.3, FR-05.6, FR-03.3, BR-03, NFR-04.
Relationships	Included by UC-08, UC-09, and UC-15.
Description	The system changes the stock of one or more products and records each stock movement.
Main flow	1. For each product, the system calculates the new stock: it subtracts the quantity for a sale and adds it for a purchase or a voided sale. 2. The system verifies that the new stock is not negative (BR-03). 3. The system updates the current stock of the product. 4. The system records a stock movement with the date and time, the product, the type, the quantity, the related sale or purchase, and the user. 5. The product appears in or disappears from the list of stock alerts according to its new stock (FR-08.4).

Exceptions and secondary paths	2a. The stock would be negative: the whole operation is rejected.

UC-12. VIEW INVENTORY	
Primary actor	Warehouse Clerk
Backlog and requirements	RF05 · FR-05.5, FR-05.6.
Relationships	None.
Description	The warehouse clerk reviews the current stock of the products and the history of stock movements of each one.
Main flow	1. The warehouse clerk opens the inventory option. 2. The system lists the active products with their current stock and minimum stock level and highlights those at or below the minimum. 3. The warehouse clerk selects a product. 4. The system shows the stock movements of the product with their date, type, quantity, related sale or purchase, user, and reason.
Exceptions and secondary paths	**2a.** The list can be filtered by name or category.

UC-13. ADJUST STOCK	
Primary actor	Warehouse Clerk
Backlog and requirements	RF05 · FR-05.4, BR-03, BR-04, AS-05.
Relationships	None.
Description	The warehouse clerk corrects the stock of a product manually, for example, to load the initial inventory or because of damaged items or differences found in a physical count.
Main flow	1. The warehouse clerk selects the product to adjust. 2. The warehouse clerk enters the quantity (positive to add units, negative to subtract them) and the reason. 3. The warehouse clerk confirms the adjustment. 4. The system adds or subtracts the quantity from the stock of the product, verifying that it does not become negative (BR-03), and records a stock movement with the type “adjustment”, the user, the date, and the reason. 5. The system shows the new stock of the product.
Exceptions and secondary paths	**3a–4a.** Empty reason, or the stock would be negative: the adjustment is rejected.

UC-14. MANAGE SUPPLIERS	
Primary actor	Owner
Backlog and requirements	RF06 · FR-06.1, FR-06.2, BR-07, BR-08.
Relationships	None.
Description	The owner registers, updates, searches for, and deactivates the suppliers of the company.
Main flow	1. The owner opens the suppliers option and selects the option to register a supplier. 2. The owner enters the RUC, the business name, the phone number, and, optionally, the contact person, the email, and the address (Table B3 of the SRS). 3. The system verifies that the RUC has 13 digits and is unique, that the required fields are complete, and that the email has a valid format. 4. The system saves the supplier as active and shows it in the supplier list.
Exceptions and secondary paths	**A1–A3.** Search by RUC or name; update the data (except the RUC); deactivate (it stays in the history). **3a.** RUC already exists or invalid field: error message.

UC-15. REGISTER PURCHASE	
Primary actor	Warehouse Clerk
Backlog and requirements	RF06 · FR-06.3, FR-06.4, FR-06.5, FR-05.2, BR-02, BR-04, NFR-04.
Relationships	«include» UC-06 Search products; «include» UC-11 Update stock.
Description	The warehouse clerk registers the goods received from a supplier; the system increases the stock.
Main flow	1. The warehouse clerk selects the option to register a new purchase; the system records the logged-in user and proposes the date of today. 2. The warehouse clerk selects an active supplier and, optionally, enters the supplier’s invoice number. 3. The warehouse clerk searches for a product (UC-06) and enters the quantity and the unit cost. 4. The system verifies that the quantity and the unit cost are greater than zero, adds the line, and updates the total cost. Steps 3 and 4 are repeated for each product. 5. The warehouse clerk confirms the purchase. 6. The system verifies that the purchase has a supplier and at least one product, assigns the next consecutive number, and saves it. 7. The system performs UC-11 Update stock to increase the stock of each product; the purchase and all its stock changes are saved together or not at all. 8. The system shows a confirmation with the number of the purchase.

Exceptions and secondary paths	1a. The date of the purchase can be changed (goods received the day before). 4a–4b. Invalid value: the line is rejected; a line can be removed and the total is recalculated. 5a–7a. The purchase is cancelled or an error occurs while saving: nothing is stored.

UC-16. VIEW SALES HISTORY	
Primary actor	Owner
Backlog and requirements	RF07 · FR-07.1, FR-07.2, FR-07.3.
Relationships	None.
Description	The owner reviews all the registered sales, filters them, and opens the details of any sale.
Main flow	1. The owner opens the sales history. 2. The system lists all the sales, including the voided ones, with their number, date, customer, salesperson, total amount, and status. 3. The owner filters the sales by date range, salesperson, customer, or status. 4. The system shows only the matching sales. 5. The owner selects a sale. 6. The system shows its products, quantities, unit prices, subtotal, VAT rate and amount, total, payment method, and, if it is voided, the reason, date, and user of the void.
Exceptions and secondary paths	**4a.** No sale matches the filters: empty list with a message.

UC-17. CHECK MINIMUM STOCK	
Primary actor	None (performed automatically by the system)
Backlog and requirements	RF08 · FR-08.1, FR-08.4, DEP-02.
Relationships	Included by UC-18 Receive stock alerts.
Description	The system decides which products must generate a stock alert.
Main flow	1. The system compares the current stock of each active product with its minimum stock level. 2. If the current stock is equal to or lower than the minimum, the system keeps the product in the list of active alerts. 3. If the current stock is higher than the minimum, the system removes the product from the list of active alerts. 4. The system updates the number of active alerts shown in the main menu.
Exceptions and secondary paths	**1a.** Deactivated products do not generate alerts.

UC-18. RECEIVE STOCK ALERTS	
Primary actor	Warehouse Clerk
Backlog and requirements	RF08 · FR-08.2, FR-08.3, UI-02, UI-06.
Relationships	«include» UC-17 Check minimum stock.
Description	The warehouse clerk sees which products need to be restocked.
Main flow	1. The system shows the number of active alerts in the main menu. 2. The warehouse clerk opens the list of alerts. 3. The system performs UC-17 Check minimum stock to obtain the list of active alerts. 4. The system shows each product at or below its minimum stock level, with its current stock, its minimum stock level, and its location.
Exceptions and secondary paths	**2a.** No active alerts: empty list with a message.

UC-19. GENERATE REPORTS	
Primary actor	Owner
Backlog and requirements	RF09 · FR-09.1, FR-09.2, FR-09.3, FR-09.5, NFR-01.
Relationships	Extended by UC-20 Export report to PDF (extension point: the report is displayed).
Description	The owner generates reports that summarize the sales, the inventory, the purchases, and the slow-moving products.
Main flow	1. The owner opens the reports option and selects the type of report: sales, inventory, purchases, or slow-moving products. 2. For the sales, purchase, and slow-moving products reports, the owner enters the start and end dates. 3. The system generates the report within 10 seconds and displays it with the content described in the notes.
Exceptions and secondary paths	**2a.** Start date later than the end date: error message. **3a.** No data in the period: the report says that there are no records.

UC-20. EXPORT REPORT TO PDF	
Primary actor	Owner
Backlog and requirements	RF09 · FR-09.4 (Could have).
Relationships	«extend» UC-19 Generate reports (extension point: the report is displayed; condition: the owner chooses to export).
Description	The owner saves a displayed report as a PDF file.

Main flow	1. The owner selects the option to export the report to PDF. 2. The owner chooses the name and the folder of the file. 3. The system creates the PDF file with the same content as the displayed report and shows a confirmation message.
Exceptions and secondary paths	3a. The file cannot be saved in the selected folder: error message.